

IT 461:
Principles of Machine Learning
Course Project
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Diamond Price Prediction

Proposal

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1. Motivation

Diamond valuation is a complex and often opaque process. The market price of a diamond is influenced by multiple factors, most prominently the “Four Cs” which are **carat**, **cut**, **color**, and **clarity**, as well as additional measurements such as depth, table size, and dimensions. Small differences in these attributes can lead to large variations in price, making it difficult for consumers and even professionals to evaluate diamonds accurately.

The problem our project aims to solve is:

Can we accurately predict the price of a diamond given its physical and categorical attributes?

This problem is important for several reasons. From a consumer perspective, a prediction model can help buyers assess whether a diamond is fairly priced. From a business perspective, jewelers and traders can use data-driven models to support valuation decisions. From an academic perspective, this dataset provides an excellent opportunity to apply regression models, categorical encoding, and performance evaluation on real-world data.

Our motivation is both practical and educational. Practically, diamonds are high-value commodities where accurate pricing matters significantly. Educationally, the project allows us to explore regression techniques, test multiple algorithms, and evaluate how preprocessing and feature engineering affect prediction performance.

1.1. **Intended Task**

The intended task of this project is to use supervised learning methods to predict diamond prices from their attributes. The input consists of features such as carat, cut, color, clarity, depth, table, and dimensions (x, y, z). The output is a numeric prediction of the diamond’s price. An example of this process is shown in Table 1, where the input features are passed through a regression model to generate a predicted price.

We will consider our project **successful** if the model achieves high predictive accuracy on unseen data. Specifically, success will be measured using standard regression metrics such as the coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE).

Input: Attributes such as carat, cut, color, clarity, depth, table, and dimensions (x, y, z).

Output: A numeric prediction of the diamond's price.

Input Features (X)		ML model Regression Model	Predicted Price (Y)
Carat	0.90		\$4,500 (approx.)
Cut	Premium		
Color	F		
Clarity	VS1		
Depth	62.0		
Table	58		
Dimensions	6.1 x 6.0 x 3.8		

Table 1. Example of the intended task: diamond attributes (input features) mapped to a predicted price using a regression model

2. Background

2.1. Key Concepts

Diamonds are among the most captivating and sought-after gems in the world, carrying a long history shaped by rarity and market influence. For decades, the diamond industry was largely controlled by monopolies that tightly managed supply to maintain high prices. In recent years, the emergence of lab-grown diamonds has disrupted this dominance, offering a more affordable and ethical alternative. Despite these changes, natural diamonds continue to hold strong emotional and cultural value, keeping their allure alive.

A diamond's price is determined by a combination of physical and visual characteristics, often summarized by the "Four Cs": cut, color, clarity, and carat. Carat measures weight, with one carat equaling 0.2 grams, and has a direct impact on price. Cut reflects how well a diamond's facets interact with light, influencing its brilliance and sparkle. Color is graded alphabetically from D (colorless) to J (lightly tinted), while clarity measures the presence of internal or external flaws, ranging from I3 (included) to FL (flawless). Other proportions, such as depth and table size, also contribute to a diamond's overall quality and value. Understanding these attributes is essential for building a machine learning model that predicts diamond prices accurately, using a dataset containing both numerical and categorical features representing these gem characteristics.

2.2. Related Works

Previous research has shown promising results in the field of diamond price prediction, highlighting the potential of machine learning models in capturing the complex relationships between a diamond's physical characteristics and its market value.

Aggarwal et al. [1] compared linear regression with decision tree algorithms and concluded that tree-based models perform better in modeling non-linear dependencies among features such as carat, cut, color, and clarity. Their findings demonstrated the limitations of simple linear models in handling the intricate interactions between diamond attributes.

Kadam and Jadhav [2] expanded upon this by applying ensemble learning techniques, including random forests and gradient boosting, which significantly improved accuracy. Their study achieved R^2 values exceeding 0.95, showing that ensemble approaches can reduce variance and capture complex data patterns more effectively than individual models.

Wang and Chen [3] explored the use of artificial neural networks (ANNs) for price prediction and observed slight performance improvements compared to traditional algorithms. However, they also noted the trade-off between increased computational complexity and model interpretability, which remains a challenge for deep learning approaches in this domain.

Community-driven analyses on Kaggle [4] have further demonstrated the value of model optimization and hyperparameter tuning. Advanced ensemble algorithms such as XGBoost and LightGBM have achieved R^2 scores approaching 0.98 when combined with proper preprocessing, encoding of categorical variables, and feature scaling techniques.

Collectively, these studies form a solid foundation for this project. Building upon prior research, our work seeks to systematically evaluate multiple regression models, perform feature engineering and selection, and analyze their comparative performance. The objective is to develop a robust, interpretable, and high-performing machine learning model capable of accurately predicting diamond prices based on their measurable characteristics.

3. Dataset

Source:

We will use the **diamonds** dataset bundled with the R package **ggplot2**, which provides ~54,000 observations linking diamond attributes (carat, cut, color, clarity, depth, table, x, y, z) to price (USD), as seen in table 2,. The dataset is **well-documented**, **widely adopted**, and **curated** as part of the ggplot2 distribution, ensuring stable access, consistent variable definitions, and reproducible research workflows [4], [5].

This dataset is particularly suitable for our problem (predicting diamond prices) because:

- Its **large sample size** supports robust training/validation splits.
- The **mix of numerical and ordinal features** reflects realistic modeling challenges (encoding, scaling, interaction effects).
- Its **established use as a benchmark** enables clear comparison with prior work and facilitates defensible methodology choices.

In short, this dataset provides the right balance of **size, feature diversity, documentation quality, and practical relevance** for building and evaluating predictive pricing models available through the official tidyverse reference page.

Key Features:

Attribute	Data Type	Description
Carat	Numerical Ratio	Carat denotes the weight of a diamond, where one carat equals 0.2 grams
Cut	Ordinal	Cut represents the diamond's body grading, ranked as: Fair, Good, Very Good, Premium, Ideal.
Color	Ordinal	Ranks the diamonds' color by assigning letters from (D-J), D being colorless and J having the most amount of color
Clarity	Ordinal	Clarity grades are ranked representing a hierarchy of quality and value. The diamond clarity scale, from worst to best, is: I3, I2, I1, SI2, SI1, VS2, VS1, VVS2, VVS1, IF, and FL
Depth	Numerical Ratio	Diamond depth is a measurement of a diamond's height, from the bottom tip to the flat top surface (table), depth is usually presented as a percentage, which is calculated by dividing the diamond's depth by its average diameter.
Table	Numerical Ratio	A diamond's table is the largest, flat facet on the top of the stone, resembling a tabletop. The size of the table is a crucial factor in a diamond's quality, and its dimensions are expressed as a table percentage, calculated by comparing the table's width to the diamond's average girdle diameter.
Price (target)	Numerical Ratio	Describes the diamond's price in USD

X	Numerical Ratio	X dimension, the longest dimension of the diamond.
Y	Numerical Ratio	Y dimension, the widest dimension of the diamond.
Z	Numerical Ratio	Z dimension, the total height or depth of the diamond.
Total # of Features		10 Predictors 1 Target
# of Distinct Classes		Cut: 5 (Fair, Good, Very Good, Premium, Ideal.) Color: 7 (D-J) Clarity: 8 (I3, I2, I1, SI2, SI1, VS2, VS1, VVS2, VVS1, IF, FL)
# of Examples		~54,000 rows

Table 2. Dataset key features, Number of classes and examples

Representative Example:

Below is a small sample of rows from the dataset to illustrate its structure:

	Unnamed: 0	carat	cut	color	clarity	depth	table	price	x	y	z
0	1	0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
1	2	0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
2	3	0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
3	4	0.29	Premium	I	VS2	62.4	58	334	4.2	4.23	2.63
4	5	0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75

4. Methods

This section describes the machine learning models used in the study, the justification for selecting each method, and the feature engineering steps applied to prepare the dataset for modeling.

4.1 Machine Learning Algorithms Employed

To capture the relationship between diamond attributes and price, we implemented a diverse set of regression models, ranging from simple **linear baselines** to **nonlinear** ensemble **methods** and neural networks.

Linear Regression: A standard regression model was used as the baseline. It assumes a linear relationship between features and price, making it useful for setting initial performance expectations and understanding feature coefficients.

Ridge Regression: Ridge adds an L2 penalty to reduce overfitting and stabilize coefficients, particularly when predictors are correlated. This model improves robustness compared to plain Linear Regression.

Lasso Regression: Lasso applies an L1 penalty, enabling feature selection by shrinking less important coefficients to zero. It helps identify the most influential predictors of price.

Random Forest Regressor: A tree-based ensemble that trains many decision trees in parallel and averages their predictions. Random Forest captures nonlinear relationships and complex interactions between diamond features.

Gradient Boosting Regressor: An ensemble of sequentially built trees, where each tree corrects errors from previous ones. Gradient Boosting is powerful for modeling subtle nonlinear effects that linear models cannot capture.

Neural Network: A feed-forward neural network with tested architectures [(64), (64,32), (128,64,32)]. Neural networks can learn deeper nonlinear patterns and interactions beyond those captured by tree models.

Final Weighted Voting Ensemble: To combine complementary strengths, the final model aggregates: Ridge, Lasso, Random Forest, Gradient Boosting, Best Neural Network (64, 32). Weights emphasize the strongest models (RF, GB, NN). The ensemble improves stability and generalization by blending linear and nonlinear learners.

4.2 Modeling Choices

Diamond pricing is influenced by highly nonlinear relationships and interactions among physical, geometric, and quality-related attributes. Therefore, multiple model classes were selected to address different aspects of the problem:

Linear Models	Linear Regression, Ridge, and Lasso were chosen because: · They provide interpretable coefficients · They establish baseline performance · They help evaluate multicollinearity and linear dependence · Ridge/Lasso offer robustness through regularization
Tree-Based Models	Random Forest and Gradient Boosting were selected because: · Diamond pricing is known to scale nonlinearly with carat · Interactions (e.g., cut $\times$ color, depth $\times$ table) are hard to model linearly · These ensembles handle skewed distributions and outliers better than linear models · They rank feature importance transparently
Neural Network	A neural network was included to explore: · Continuous nonlinear transformations · Layered feature interactions · Performance beyond tree-based models
Final Ensemble	The Weighted Voting Regressor was used because: · No single model consistently outperformed all others in every metric · Combining models reduces the risk of overfitting · Tree-based + neural + linear models together balance bias and variance

4.3 Feature Engineering and Dimensionality Reduction

1. **One-Hot Encoding of Categorical Features:** Categorical attributes (cut, color, clarity) were transformed into one-hot encoded vectors.
2. **Scaling of Numerical Features:** Numerical variables (carat, depth, table, x, y, z) were standardized using StandardScaler to normalize magnitude differences, improve convergence for linear and neural models and ensure fair contribution of all features. The target variable, price, was also scaled during training and inverse-transformed for final predictions.
3. **Outlier Handling & Data Cleaning:** - Diamonds with invalid geometries (zero dimensions) were removed and redundant indexing columns were dropped.
4. **Dimensionality Reduction:** No dimensionality reduction was used. This decision was intentional because interpretability of gemological factors (carat, cut, color, clarity) must remain intact and domain-specific meaning would be lost if compressed into latent components.

5. Experiment

5.1 Training

The training process followed a structured workflow to ensure that the models learned effectively from the diamond dataset. We began with data cleaning, removing entries with invalid or physically impossible measurements. Next, we performed feature engineering, which included one-hot encoding all categorical attributes (cut, color, clarity) and standardizing numerical features using StandardScaler. Standardization was important because several models used particularly linear regression, Ridge, Lasso, and the neural network are sensitive to differences in feature scale.

To support reliable model development, the dataset was divided into training, validation, and test sets. The training set was used to fit each model, the validation set guided architectural selection for the neural network and helped avoid overfitting, and the final test set provided an unbiased evaluation of model performance.

To improve generalization on underrepresented high-value diamonds, we applied a targeted data augmentation strategy, oversampling synthetic samples from the top 20% of diamond prices. Additionally, regularization techniques were incorporated within Ridge and Lasso regression, helping stabilize linear models by penalizing overly large coefficients.

Each model was evaluated using MAE, RMSE, and R^2 , computed after reversing the target scaling to return predictions to real dollar amounts. This ensured that the reported performance

reflected meaningful, real-world interpretation. Overall, this training and evaluation pipeline ensured fairness, consistency, and robustness across all models tested.

5.2 Architecture and Structure

In our project, we employed a combination of linear models, tree-based ensemble models, and a neural network, each offering a different architectural approach to learning patterns in diamond pricing. The linear models used Ridge Regression and Lasso Regression are built on a single weighted linear layer that maps the input features directly to the predicted price. Their key components are the feature coefficients and a regularization term that helps control overfitting. These models give us a clear, interpretable baseline and capture broad linear trends in the data.

The tree-based ensemble models, Random Forest and Gradient Boosting, provide a more flexible architecture capable of modeling nonlinear relationships. A Random Forest consists of many decision trees trained independently on random subsets of the data. Its main parameters include the number of trees, the maximum depth, and the number of features considered at each split. By averaging predictions across trees, the model reduces variance and learns complex feature interactions. Gradient Boosting builds trees sequentially, where each new tree focuses on correcting the errors of the previous ones. Its learning rate, number of boosting iterations, and tree depth determine how effectively it captures subtle patterns while preventing overfitting.

Finally, our neural network model uses a multilayer perceptron architecture with two hidden layers of 64 and 32 nodes. Each layer employs a ReLU activation function, allowing the network to learn nonlinear relationships by transforming the feature space at each layer. Through backpropagation, the network adjusts its internal weights to extract deeper and more abstract patterns from the data. Together, these architectures give our project a balanced mix of interpretability, robustness, and expressive modeling power, enabling the system to handle both simple and complex relationships in the diamond dataset.

5.3 Hyperparameters

To optimize model performance and ensure strong generalization, a structured hyperparameter-tuning procedure was applied to all machine learning models. Grid Search Cross-Validation was used as the search strategy, as it systematically evaluates all combinations of predefined hyperparameters and is suitable for the scale of the dataset. For the Ridge regression model, the regularization strength parameter (α) was tuned over the range $\{0.01, 0.1, 1, 10\}$, allowing the model to balance bias and variance while addressing multicollinearity among correlated features. Random Forest tuning focused on parameters that directly affect model complexity and variance reduction, including the number of trees (100, 200, 300), maximum tree depth (None, 10, 20), and minimum samples required to split a node (2, 5). These ranges

allowed the search to compare shallow, moderately deep, and fully grown trees, ensuring the final configuration captured nonlinear patterns without overfitting. For the Gradient Boosting model, the learning rate (0.01, 0.05, 0.1), number of boosting stages (100, 200), and tree depth (2, 3) were optimized. These values reflect commonly effective ranges that balance weak learner complexity with boosting stability.

Across all models, Grid Search automatically trained each hyperparameter combination using cross-validation folds and selected the configuration that produced the lowest average validation error, with RMSE serving as the primary selection metric. The models were then retrained on the full training set using the best-identified parameters. This tuning process ensured that the final models were neither underfit nor overfit, and that their performance remained stable and robust when evaluated on unseen data.

5.4 Performance Metrics

The following metrics were used to evaluate model performance:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

5.5 Python Libraries and Resources

The entire workflow was executed in **Google Colab**, Jupyter notebook services.

Python Libraries used:

- pandas
- numpy
- scikit-learn
 - LinearRegression – baseline linear regression model.
 - Ridge, Lasso – regularized linear models.
 - RandomForestRegressor – tree-based ensemble model.
 - GradientBoostingRegressor – boosting-based tree ensemble.
 - MLPRegressor – neural network model.
 - VotingRegressor – final weighted ensemble model.
 - GridSearchCV – hyperparameter tuning for Ridge, Lasso, RF, and GB.
 - train_test_split – dataset splitting.
 - KFold, cross_val_score – cross-validation.
 - StandardScaler – feature scaling for inputs and outputs.
- matplotlib / seaborn (for EDA)

Resources:

The notebook was executed on cloud services and does not require any particular hardware other than a laptop.

6. Results and Discussion

6.1 Model Performance

The following table demonstrates the resulted metrics from each trained model:

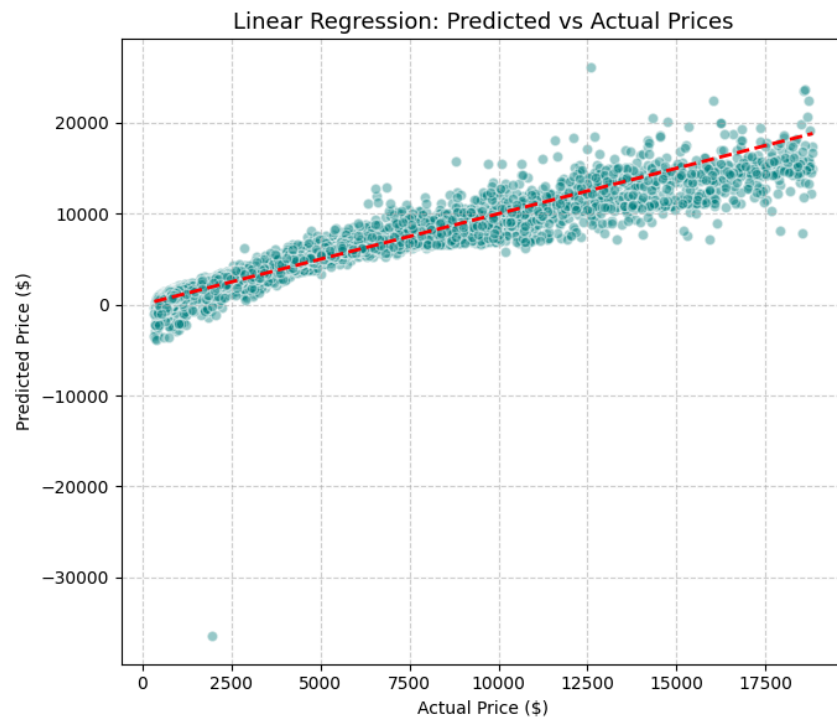
	MAE	RMSE	R^2
Ridge Model	721.03	1,137.33	0.9193
Lasso Model	718.75	1,126.77	0.9208
Random Forest	296.84	613.19	0.9765
Gradient Boosting	320.39	625.31	0.9793
Neural Network	307.15	576.63	0.9793
Ensemble	318.71	578.19	0.9791

6.2 Model Performance on Unseen Data

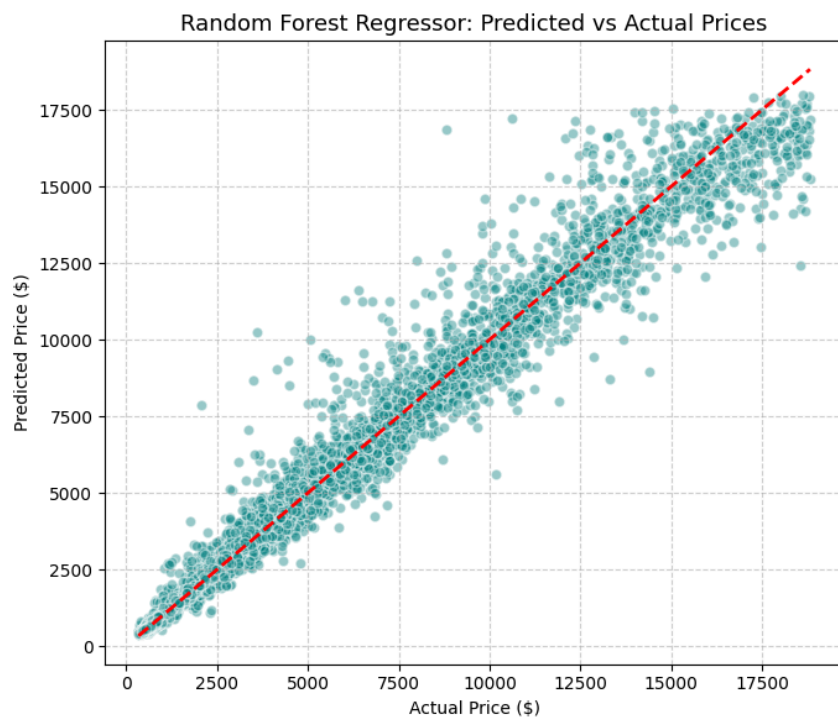
Overall, the models demonstrated strong generalization to unseen data, as shown by their performance on the test set. Linear models such as Ridge and Lasso achieved reasonable R^2 score 0.92, indicating that they captured the general trend but struggled with the nonlinear structure of diamond pricing. In contrast, the tree-based models Random Forest and Gradient Boosting generalized well, achieving R^2 values above 0.97 with low MAE and RMSE, showing that they effectively learned complex relationships without overfitting. The neural network also generalized strongly, matching the tree-based models with an R^2 of 0.979, demonstrating stable performance across different price ranges. Although the Voting Ensemble did not surpass the best individual models, it delivered consistently high accuracy and stable predictions, confirming that combining multiple learning strategies enhances robustness. The close alignment between

training, validation, and test scores across all advanced models indicates that the system generalizes well and can make reliable predictions on real-world unseen diamonds.

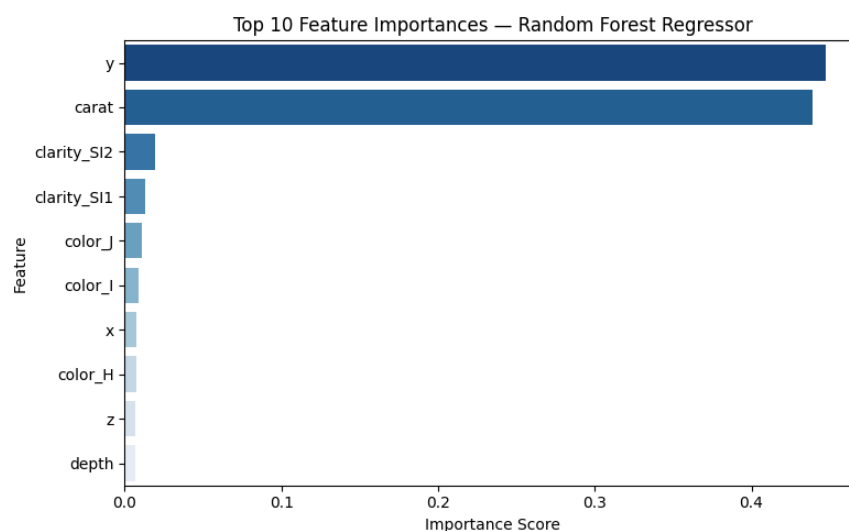
6.3 Visualization



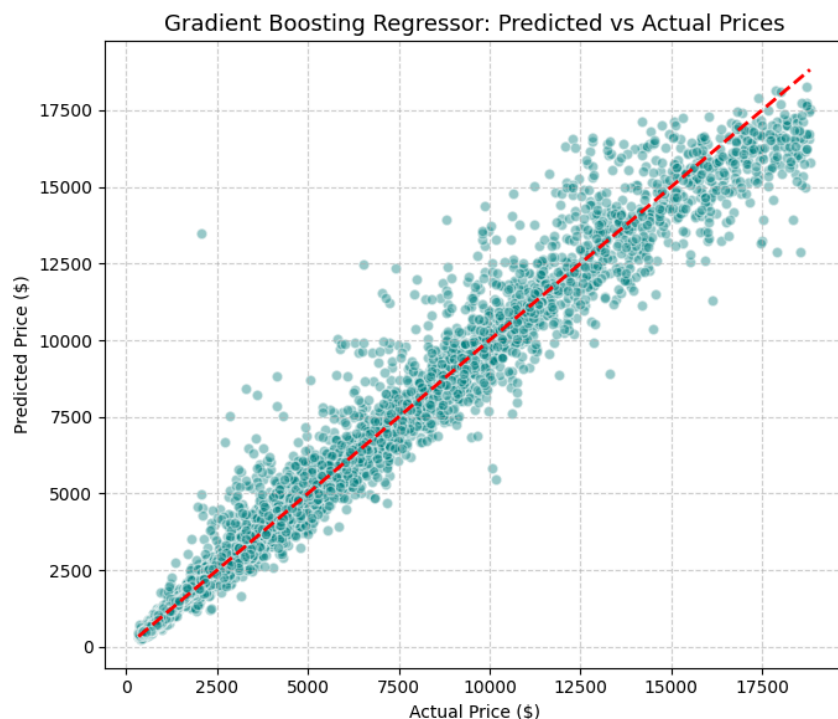
The scatter plot illustrates how the Linear Regression model's predicted diamond prices compare to the actual prices in the test set.



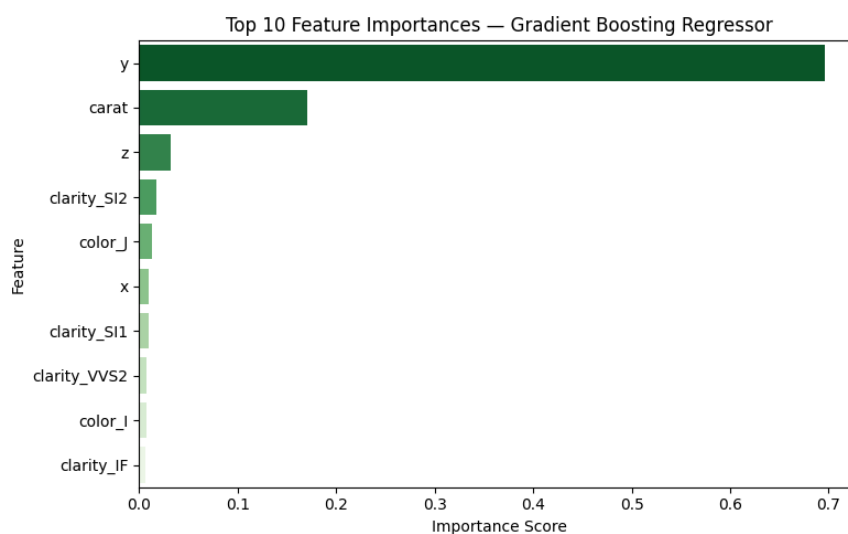
The scatter plot illustrates how the Random Forest model's predicted diamond prices compare to the actual prices in the test set.



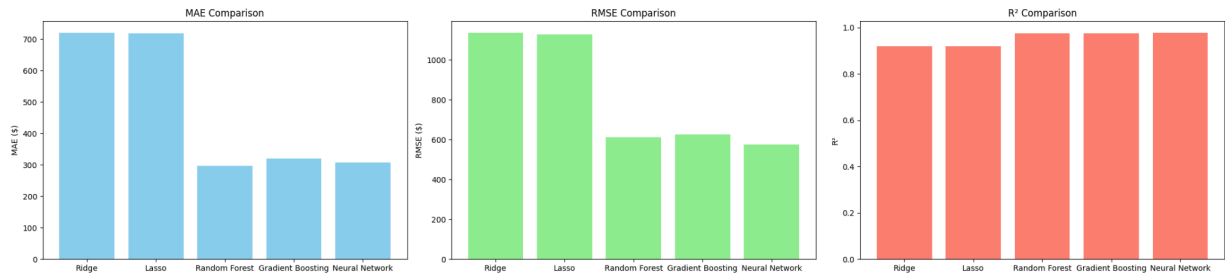
This bar chart highlights the ten most influential features used by the Random Forest Regressor when predicting diamond prices.



The scatter plot illustrates how the Gradient Boosting model's predicted diamond prices compare to the actual prices in the test set.



This bar chart highlights the ten most influential features used by the Gradient Boosting Regressor when predicting diamond prices.



The bar chart illustrates the performance of all five models-Ridge, Lasso, Random Forest, Gradient Boosting, and the Neural Network-by comparing their MAE, RMSE, and R^2 scores. This visualization provides a clear overview of how each algorithm handles the diamond pricing task, highlighting the superior accuracy of the tree-based and neural network models compared to linear approaches.

6.4 Findings and Insight

The results of our analysis provide several meaningful insights into both the dataset and the behavior of the predictive models. First, the models consistently confirmed that carat weight is the single strongest driver of price, followed by the diamond's physical dimensions (x, y, z). These findings align with real market behavior, where size heavily influences value. Quality factors such as clarity, color, and cut still contributed to price variation, but their impact was noticeably smaller compared with physical characteristics.

The experiments also revealed that diamond pricing is highly nonlinear. Linear models like Ridge and Lasso performed adequately but were unable to fully capture the complex interactions present in the data. In contrast, tree-based models and the neural network achieved significantly higher accuracy, indicating that nonlinear relationships play a major role in determining price. The Random Forest and Gradient Boosting models excelled by identifying subtle feature interactions, while the neural network further improved performance by learning deeper nonlinear patterns.

An interesting finding is that the Voting Ensemble, although not the top performer in any single metric-offered one of the most stable and balanced prediction profiles. This shows that blending different modeling strategies can yield a reliable, general-purpose predictor even when individual models specialize in different aspects of the data.

Overall, the analysis highlights that effective diamond price prediction requires models capable of handling nonlinearities, feature interactions, and varying scales. The combination of traditional regression

models, tree-based ensembles, and neural networks provided a clear picture of the strengths and weaknesses of each approach, ultimately revealing how different algorithms interpret and respond to the underlying structure of diamond pricing.

7. Conclusion

7.1 Key Findings

Our analysis revealed several important patterns in real diamond pricing. Carat weight is by far the strongest driver of price, and even small increases in size lead to substantial jumps in value. Dimensions (x, y, z) also play a major role, showing that the physical volume of a diamond influences price more than any single quality grade. While factors like clarity, color, and cut do affect price, their impact is more moderate compared to carat and overall size. We also found that high-priced diamonds are rarer and more variable, which explains why they are harder to predict consistently.

On the modeling side, models that capture nonlinear relationships-such as Random Forest, Gradient Boosting, and Neural Networks-performed significantly better than linear methods. The neural network achieved the highest accuracy, while the voting ensemble provided the most stable overall performance. These results highlight that combining multiple modeling strategies and incorporating detailed size-based features leads to the most reliable diamond price predictions.

Overall, the project confirms that diamond pricing is shaped by both physical characteristics (especially size) and quality grades, and that accurate price estimation requires models capable of learning complex interactions rather than simple linear trends.

7.2 Challenges and Limitations

One of the main challenges we faced was the uneven distribution of diamond prices, especially the scarcity of very high-priced diamonds. This made it harder for models to learn consistent patterns for the upper price range, even after applying targeted oversampling. Another limitation is that our dataset did not include several real-world factors-such as diamond shape, certification type, fluorescence intensity, or market conditions-all of which influence price in practice. Although the models performed well, they remain sensitive to outliers and to the natural variability of rare diamonds. Additionally, the neural network and tree-based models required careful tuning and were computationally more demanding, which may affect scalability. Finally, because our dataset reflects a single point in time, the models do not

account for changing market trends, meaning the predictions may degrade without periodic retraining on updated data.

7.3 Future Directions

Future work could explore larger multi-source datasets, integrate richer diamond attributes such as shape, certification details, fluorescence, and polish, and include distinctions between natural and lab-grown diamonds. More advanced methods such as deep learning architectures, automated feature extraction, or time-aware pricing models could also be investigated.

8. Contributions

Name	Task & Contribution
Jood Alkhrashi	Contributed by writing the motivation section. Developed and deployed the Neural Network model and contributed to building the final ensemble.
Najla Almazyad	Prepared the key concept section. Implemented and optimized the tree-based models including Random Forest and Gradient Boosting.
Nouf Almansour	Collaborated on searching and documenting the dataset section. Implemented the Linear Regression models.
Sara Aloqiel	Collaborated on searching and documenting the dataset section. Conducted Exploratory Data Analysis.
All Members	Collaboratively discussed findings, interpreted results, and jointly worked on writing and refining the project report.

9. References

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